Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	
Team ID	Team 12
Project Name	Smart library request workflow
Maximum Marks	4 Marks

Technical Architecture:

The **Smart Library Request Workflow** uses cloud-based services to process user requests submitted via audio or text, converting them into structured data using speech recognition and natural language understanding.

The processed data is stored in a database and accessed

Order processing during pandemics for offline mode:

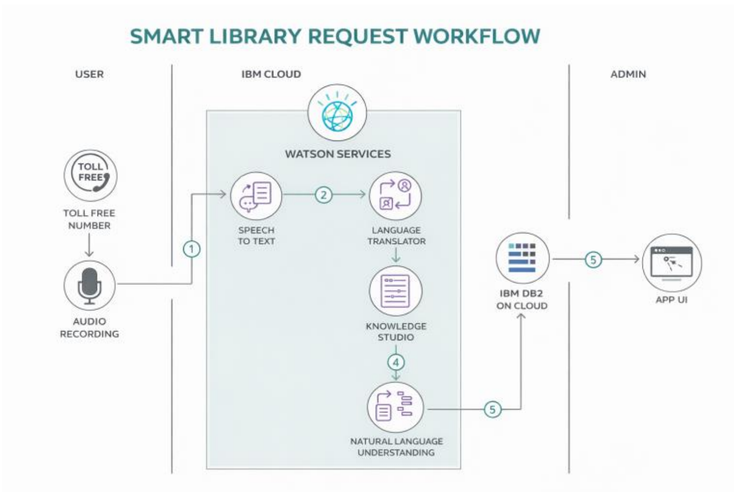

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for users to request and track books	HTML, CSS, JavaScript / React / Angular
2.	Application Logic-1	Handles request processing and user management	Python / Java
3.	Application Logic-2	Converts speech to text	IBM Watson STT
4.	Application Logic-3	Processes user queries intelligently	IBM Watson Assistant
5.	Database	Stores users, books, and requests	MySQL / MongoDB
6.	Cloud Database	Cloud-based data storage	IBM DB2 / Cloudant
7.	File Storage	Stores files and documents	Cloud Storage / Local
8.	External API-1	Fetches book details	. Google Books API
9.	External API-2	Provides notifications/verification	SMS / Email API
10.	Machine Learning Model	Recommends books	ML Model (Python)
11.	Infrastructure (Server / Cloud)	Deploys application	Cloud / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Used for building UI and backend services	React JS, Node.js, Flask

S.No	Characteristics	Description	Technology
2.	Security Implementations	Ensures secure data access and user authentication	HTTPS, SHA-256, JWT, IAM Controls
3.	Scalable Architecture	Supports increasing users using modular design	Microservices, Cloud (IBM Cloud / AWS)
4.	Availability	Ensures system is always accessible	Load Balancer, Cloud Servers
5.	Performance	Improves speed and handles multiple requests efficiently	Caching, CDN, Optimized APIs